

35. Keyboard: keydown aur keyup

Input field ke kisi bhi change ko pakadna ho (paste, speech, mouse) to keyboard events kaafi nahi hain, `input` event use karo (agla topic). Keyboard events tab use karo jab **keyboard ki actions** handle karni hon (arrow keys, hotkeys).

Events

- `keydown`: key dabai (lambi der dabao to **auto-repeat**, baar baar `keydown`)
- `keyup`: key chhodi

Auto-repeat wale events me `event.repeat === true`.

`event.key` VS `event.code`

- `event.key`: character (jaise `z` ya `Z`), language/case ke saath badalta hai.
- `event.code`: **physical key** ka code, hamesha same.

Key	<code>key</code>	<code>code</code>
<code>Z</code>	<code>z</code>	<code>KeyZ</code>
<code>Shift+Z</code>	<code>Z</code>	<code>KeyZ</code>
<code>F1</code>	<code>F1</code>	<code>F1</code>
<code>Shift</code>	<code>Shift</code>	<code>ShiftLeft</code> / <code>ShiftRight</code>

Codes: letters `KeyA`..., digits `Digit0`..., special: `Enter`, `Backspace`, `Tab`. Case matters: `"KeyZ"` sahi, `"keyZ"` galat.

Kaun sa use karein?

- Hotkey jo language switch par bhi chale** -> `event.code`

```
js
document.addEventListener('keydown', function(event) {
  if (event.code == 'KeyZ' && (event.ctrlKey || event.metaKey)) alert('Undo!');
});
```

- Layout-dependent character** chahiye -> `event.key`

Dhyan: `event.code` alag layouts me alag character de sakta hai. German layout (QWERTZ) me `Y` dabane par `code = KeyZ` aata hai. (Ye kuch hi codes ke saath hota hai: `KeyA`, `KeyQ`, `KeyZ` jaise.)

Default actions

`keydown` par kai cheezein hoti hain: character aana, delete, page scroll (`PageDown`), `Ctrl+S` save dialog. `keydown` me `preventDefault()` se zyada tar roke ja sakte hain. (OS-level keys jaise Windows me `Alt+F4` nahi rok sakte.)

Example, phone input me sirf digits aur `+ ( ) -`:

```
js
function checkPhoneKey(key) {
  return (key >= '0' && key <= '9') ||
    ['+', '(', ')', '-', 'ArrowLeft', 'ArrowRight', 'Delete', 'Backspace'].includes(key)
}
// <input onkeydown="return checkPhoneKey(event.key)" type="tel">
```

`return false` (DOM property / attribute se) default action rokta hai. Filter me `Backspace`, arrows jaise special keys bhi allow karne padte hain, warna wo bhi kaam nahi karenge.

Ye filter 100% bharosemand nahi: mouse se paste ya mobile ke dusre input tarike bach nikalte hain. Behtar: `oninput` se **baad me** value check karo (ya dono use karo).

Legacy (mat use karo)

`keypress` event aur `keyCode`, `charCode`, `which` properties purani hain, browsers me inconsistent thin, ab deprecated.

Mobile keyboards (IME)

Virtual keyboards par standard ke hisaab se `keyCode = 229` aur `key = "Unidentified"` ho sakta hai. Kuch keys (arrows, backspace) par sahi values aati hain, par guarantee nahi, to mobile par keyboard logic bharosemand nahi.

Summary

- Har key (Shift, Ctrl bhi) keyboard event deti hai. Sirf `Fn` key ka event nahi aata.
- `keydown` (auto-repeat), `keyup`.
- `code` = physical key, `key` = character.
- Form fields ke input ke liye `input`/`change` events use karo, keyboard events tab jab keyboard hi chahiye (hotkeys, special keys).

Task

Extended hotkeys (`runOnKeys(func, "KeyQ", "KeyW")`): ek `Set` me abhi dabi hui keys rakho. `keydown` par add, `keyup` par remove. Har `keydown` par check karo ki saari zaruri keys set me hain to `func` chalo.